Random Forest Classification using Mushroom Dataset

1. Introduction

Machine learning plays an important role in solving real-world classification problems by identifying patterns in structured datasets. In this project, the **Random Forest Classification algorithm** is applied to the **Mushroom Dataset** to classify mushrooms as either **edible** or **poisonous** based on their physical and biological characteristics.

Random Forest is a supervised ensemble learning algorithm that builds multiple decision trees and combines their predictions to improve classification accuracy and reduce overfitting. Because of its robustness and ability to handle complex datasets, it is widely used for classification tasks.

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2. Aim

The aim of this project is to build a **Random Forest Classification model** using the **Mushroom dataset** and evaluate its performance in accurately classifying mushrooms as edible or poisonous.

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3. Objectives

The main objectives of this project are:

- To read the dataset from a CSV file
- To understand the dataset structure
- To perform exploratory data analysis
- To encode categorical data into numerical format
- To train a Random Forest Classifier
- To evaluate the model performance
- To analyze feature importance

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4. Tools and Libraries Used

This project was implemented using **Python** and several important libraries for data analysis, visualization, machine learning, and model evaluation.

The libraries used include:

- **pandas** for data handling and manipulation
- **numpy** for numerical operations
- **matplotlib** and **seaborn** for data visualization
- **scikit-learn** for model building, preprocessing, and evaluation

These libraries made it possible to carry out the complete workflow of the project efficiently, from loading the dataset to evaluating the final model.

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5. Dataset Description

The dataset used in this project is the **Mushroom Dataset**, which was read from a CSV file. It contains **61,069 rows** and **21 columns**. Each row represents one mushroom sample, and the target variable is **class**, which indicates whether a mushroom is:

- **e** = edible
- **p** = poisonous

The dataset contains several mushroom characteristics such as:

- cap diameter
- cap shape
- cap surface
- cap color
- gill attachment
- gill spacing
- gill color
- stem height
- stem width
- habitat
- season

Some columns contain numerical values, while most of the features are categorical.

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6. Data Exploration

The dataset was initially explored using methods such as:

- `head()` to view the first few rows
- `describe()` to generate a statistical summary
- `value_counts()` to check class distribution

The statistical summary showed that the dataset contains numerical values such as **cap diameter**, **stem height**, and **stem width**, with no major structural issues.

The class distribution showed that the dataset contains:

- **33,888 poisonous mushrooms**
- **27,181 edible mushrooms**

This indicates that the dataset is relatively balanced, which is beneficial for training a reliable classification model.

A **class distribution plot** was also created to visually represent the number of edible and poisonous mushrooms in the dataset.

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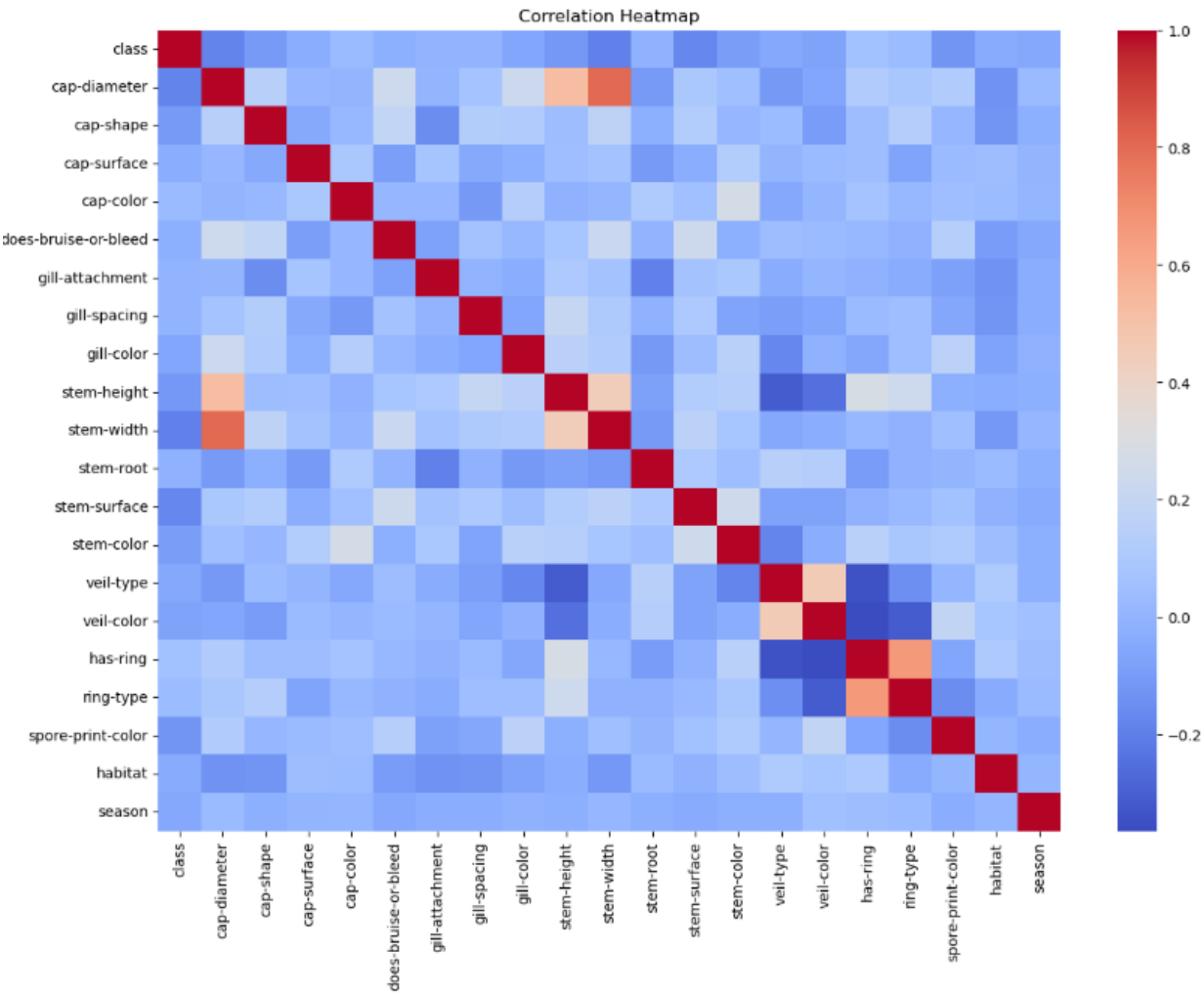
7. Data Preprocessing

Since the Mushroom dataset contains many categorical features represented by letters and symbols, it was necessary to convert them into numerical format before applying the machine learning model.

For this purpose, **Label Encoding** was used. Each categorical value in the dataset was transformed into a corresponding numerical label. This step is essential because machine learning algorithms such as Random Forest cannot directly process text-based categories.

After encoding, all columns in the dataset were converted into numerical values, making the dataset suitable for model training.

A **correlation heatmap** was also generated to study the relationships among the encoded features. This provided useful insight into how strongly different mushroom characteristics are related to one another.



8. Feature and Target Selection

The dataset was divided into two parts:

- **X (Features):** All input variables used for prediction
- **y (Target Variable):** The `class` column

Here:

- **X** contains the mushroom characteristics
- **y** contains the class label indicating whether the mushroom is edible or poisonous

This separation is a necessary step before training any supervised machine learning model.

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9. Splitting the Dataset

To evaluate the performance of the model properly, the dataset was split into **training data** and **testing data**.

- **80%** of the dataset was used for training
- **20%** of the dataset was used for testing

The final split sizes were:

- **Training Data Shape:** (48,855, 20)
- **Testing Data Shape:** (12,214, 20)

This step ensures that the model is tested on unseen data rather than only memorizing the training samples.

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10. Model Building

The classification model used in this project is the **Random Forest Classifier**.

The model was initialized with the following main parameters:

- `n_estimators = 100`
- `random_state = 42`

This means the algorithm builds **100 decision trees**, and the fixed random state ensures that the results remain consistent every time the code is executed.

The model was then trained using the training dataset.

In addition, the model was also tested with **Out-of-Bag (OOB) scoring** enabled:

- `oob_score = True`

Out-of-Bag scoring is an internal validation technique used in Random Forest. During training, each tree is built using a random subset of the training data, and the remaining unused samples are used to estimate model performance. This provides an additional validation mechanism without requiring a separate validation dataset.

The model also showed that the Random Forest successfully created **100 decision trees**.

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11. Prediction and Evaluation

After training, the model was used to make predictions on the testing dataset.

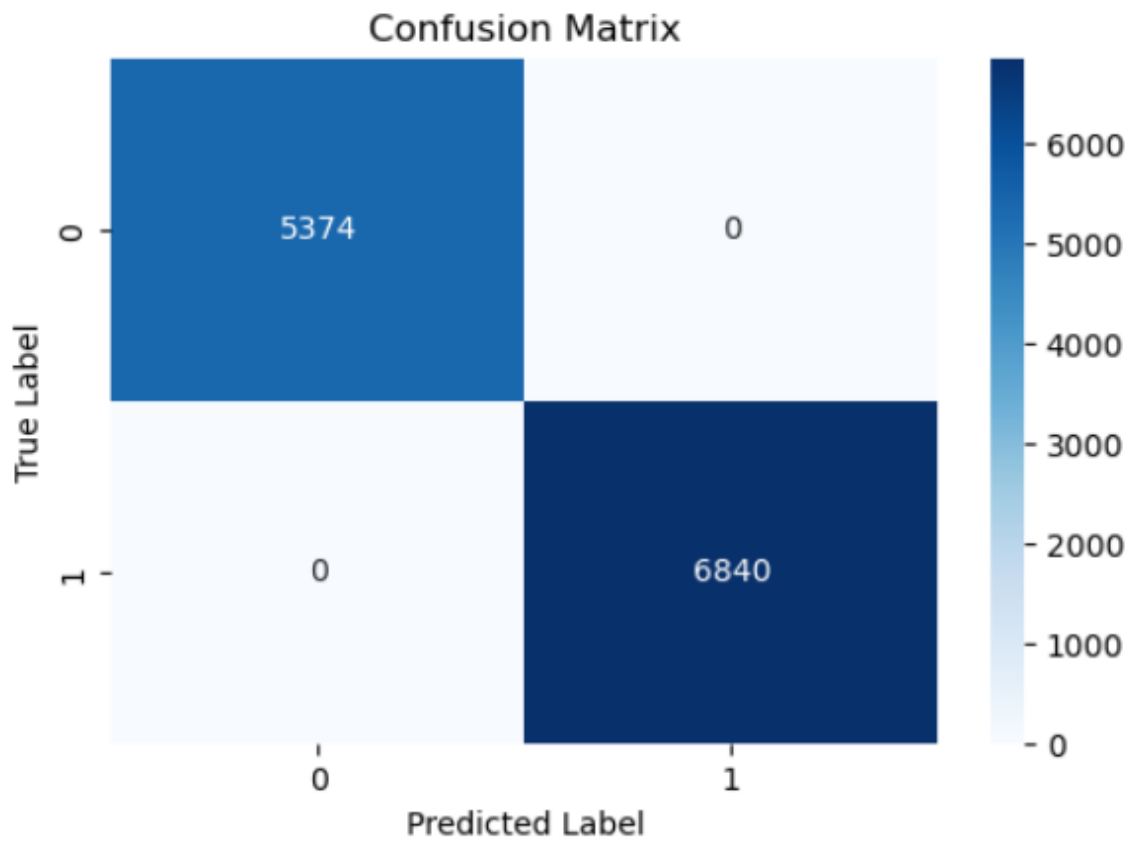
The evaluation results were extremely strong:

- **Model Accuracy:** `1.0`
- **Training Accuracy:** `1.0`
- **Testing Accuracy:** `1.0`
- **Out-of-Bag Score:** `0.9999`

These values indicate that the model performed exceptionally well on this dataset.

Confusion Matrix

A confusion matrix was generated to analyze the number of correct and incorrect predictions. The matrix showed that the model classified the mushrooms with **no misclassifications**, which confirms its perfect prediction performance on the test data.



Classification Report

A classification report was also generated, which included:

- **Precision**
- **Recall**
- **F1-score**
- **Support**

For both classes, the results were:

- **Precision:** `1.00`
- **Recall:** `1.00`
- **F1-score:** `1.00`

This confirms that the Random Forest model was highly accurate and reliable in distinguishing between edible and poisonous mushrooms.

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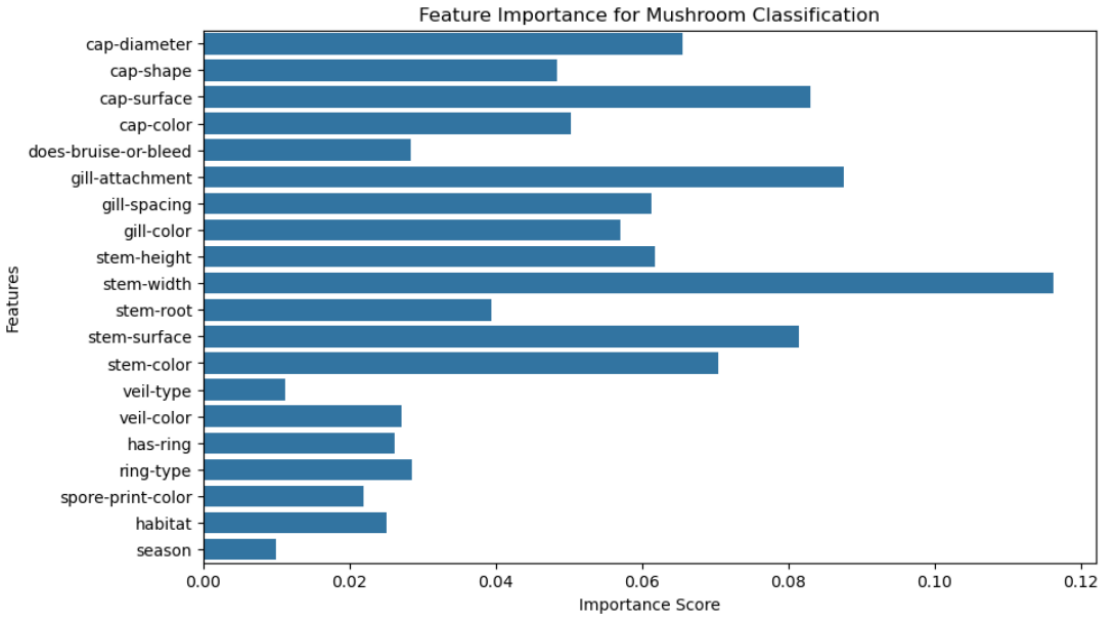
12. Feature Importance Analysis

One of the most useful advantages of Random Forest is that it can measure **feature importance**.

In this project, two methods were used:

1. Built-in Feature Importance

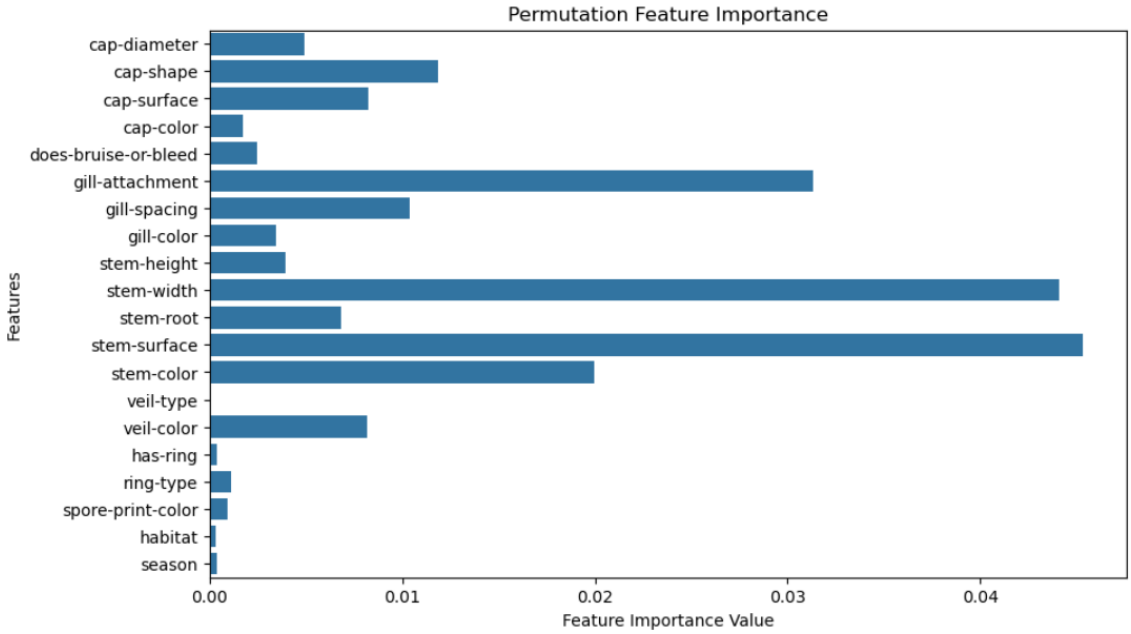
This method identifies which mushroom characteristics contribute the most to the classification process.



2. Permutation Importance

This method works by randomly shuffling one feature at a time and checking how much the model's performance decreases. If the accuracy drops significantly, that feature is considered important.

These analyses helped in understanding which mushroom features played the biggest role in prediction and made the model more interpretable.



Accuracy Comparison Plot

Description

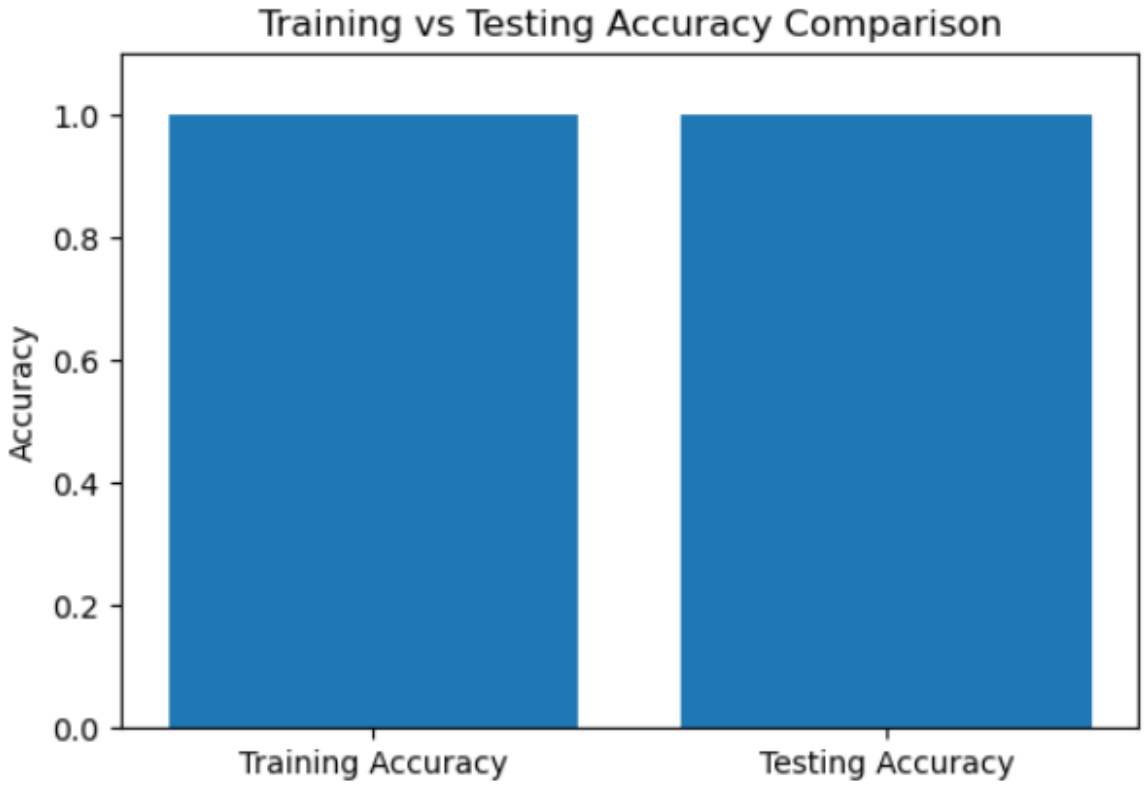
This plot is used to visually compare the **training accuracy** and **testing accuracy** of the Random Forest model.

Comparing these two values helps in checking whether the model is:

- **well-generalized**
- **overfitting**
- **underfitting**

If both values are close to each other, it usually means that the model is performing consistently on both seen and unseen data.

In this project, both training and testing accuracy are **1.0**, which indicates that the model has performed extremely well on the Mushroom dataset.



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14. Conclusion

In this project, the **Random Forest Classification algorithm** was successfully applied to the **Mushroom Dataset** to classify mushrooms as **edible** or **poisonous**.

The dataset was properly explored, preprocessed, encoded, and divided into training and testing sets. The Random Forest model was then trained and evaluated successfully.

The model achieved:

- perfect testing accuracy
- excellent classification performance
- strong internal validation through Out-of-Bag scoring
- meaningful feature importance analysis

Final Conclusion

This project clearly demonstrates that **Random Forest** is a highly effective machine learning algorithm for structured classification problems. It provides:

- high predictive accuracy

- strong generalization ability
- resistance to overfitting
- useful interpretability through feature importance

Overall, the project was successful in both **technical performance** and **analytical understanding**, making it a strong example of a classification-based machine learning application.

References

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Github Link:

https://github.com/Jahnavi-1305/Machine_learning-Individual-Project.git